CHECKPOINT ASP.NET CORE WEB APPLICATION

MVC Architecture, Cookie Security & Directory Operations

A comprehensive breakdown of multi-layer web architecture, EF Core database management, Cookie Authentication, and full CRUD workflows.

ASP.NET Core 10 MVC **Cookie Authentication** **EF Core & SQLite**

SYSTEM ARCHITECTURE

Core Technology Stack

ASP.NET Core MVC

Clean separation of Concerns with Models, Razor Views, and Controllers.

EF Core & SQLite

ORM database mapping with LINQ query execution.

Cookie Auth

Stateful session security with Encrypted Claims Principal.

Bootstrap 5 UI

Responsive layout forms, alerts, and table styles.

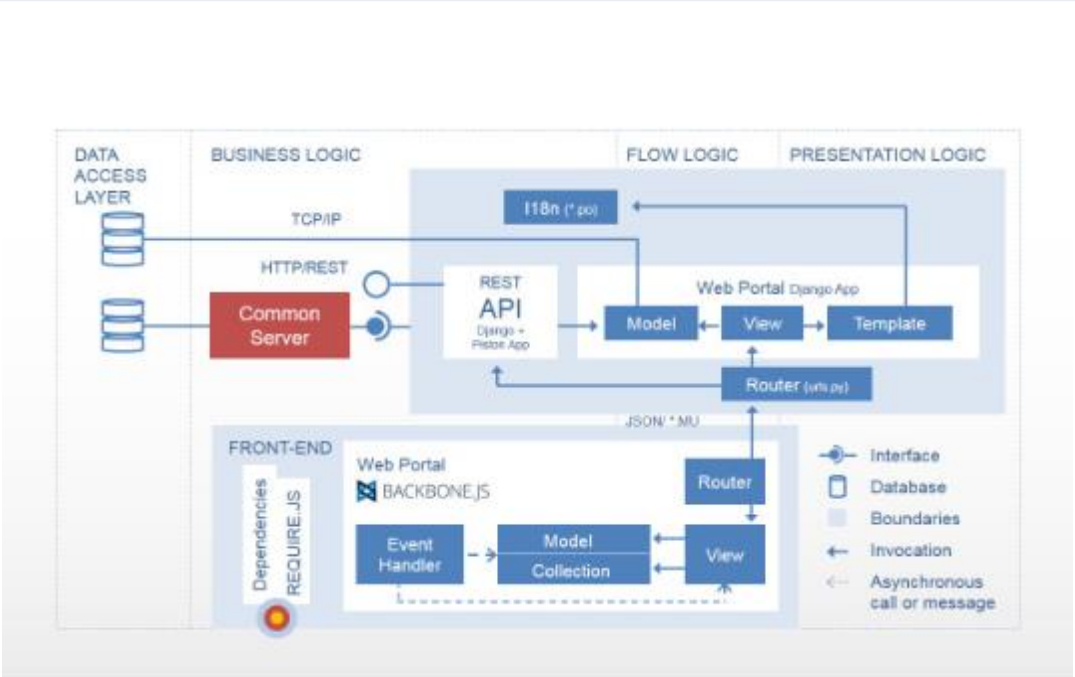
DESIGN PATTERN

Model-View-Controller Layering

Models (Data Structure)
Entity definitions for User credentials and Student directory attributes.

Views (Presentation)
Dynamic Razor templates (.cshtml) rendering responsive UI controls.

Controllers (Application Logic)
HTTP request handlers directing data processing and view routing.



PERSISTENCE LAYER

Data Models & Entity Framework Core

User Entity

Stores registered account profile and login credentials.

```
EmailId, Password, Address, ContactNo
```

Student Entity

Stores directory record details for academic tracking.

```
FirstName, LastName, Email, Course
```

ApplicationDbContext

Manages SQLite database connections and migrations.

```
DbSet, DbSet
```

AUTHENTICATION ARCHITECTURE

Cookie-Based Security Pipeline



Claims-Based Identity

Stores User ID and Email inside secure ClaimsPrincipal payload.

Encrypted Cookie Storage

Issues HttpOnly encrypted session cookies upon successful sign-in.

[Authorize] Attribute

Protects StudentController routes from unauthenticated access.

APPLICATION SECURITY

Defense Mechanisms & Validation

CSRF Protection

[ValidateAntiForgeryToken] defends POST endpoints against cross-site forged requests.

Model Validation

DataAnnotations like [Required] and [EmailAddress] ensure server-side data integrity.

Client Scripts

jQuery Unobtrusive validation provides immediate UI feedback without page reloads.

WORKFLOW DEMO

User Authentication Pipeline

01. Register

Create User

Submits account profile to database.

02. Verify

Login Post

Checks email & password in EF Core.

03. Authorize

Issue Cookie

Signs in ClaimsPrincipal instance.

04. Redirect

Student Index

Grants access to protected views.

CORE FEATURE

Student Directory Query Engine

Multi-Column Search

Filters dynamically across Name, Email, and Course major via LINQ.

Dynamic Sorting

Toggles column sorts between ascending and descending order.

Bootstrap Table UI

Clean table layout with hover states and action buttons.

DATA MANAGEMENT

CRUD Operations & TempData Notifications

Controller Operations

Create: Inserts valid Student entity to EF Core.

Edit: Fetches existing student record by ID and updates details.

Delete: Removals safeguarded with confirmation prompts.

User Alerts (TempData)

Success Banners: Triggers dynamic alert popups upon saving.

Error Banners: Displays validation failure notifications.

Auto-Dismiss: Dismissible Bootstrap alert components.

PROJECT COMPLETE

Summary & Live Demo Q&A

The CheckPoint application demonstrates clean MVC separation, secure Cookie Authentication, robust input validation, and full CRUD capability.

Repo: CheckPoint2App

Ready for Presentation